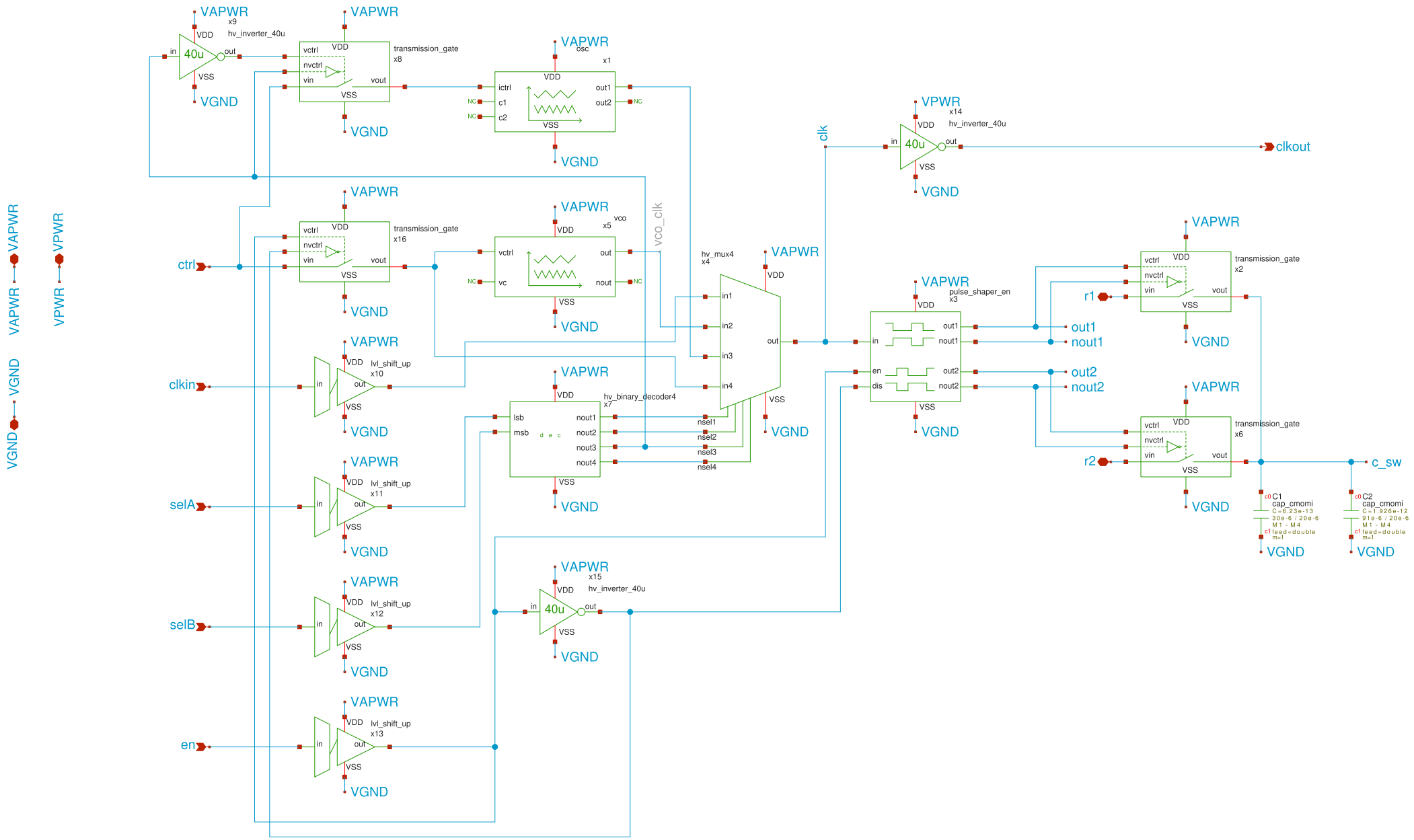
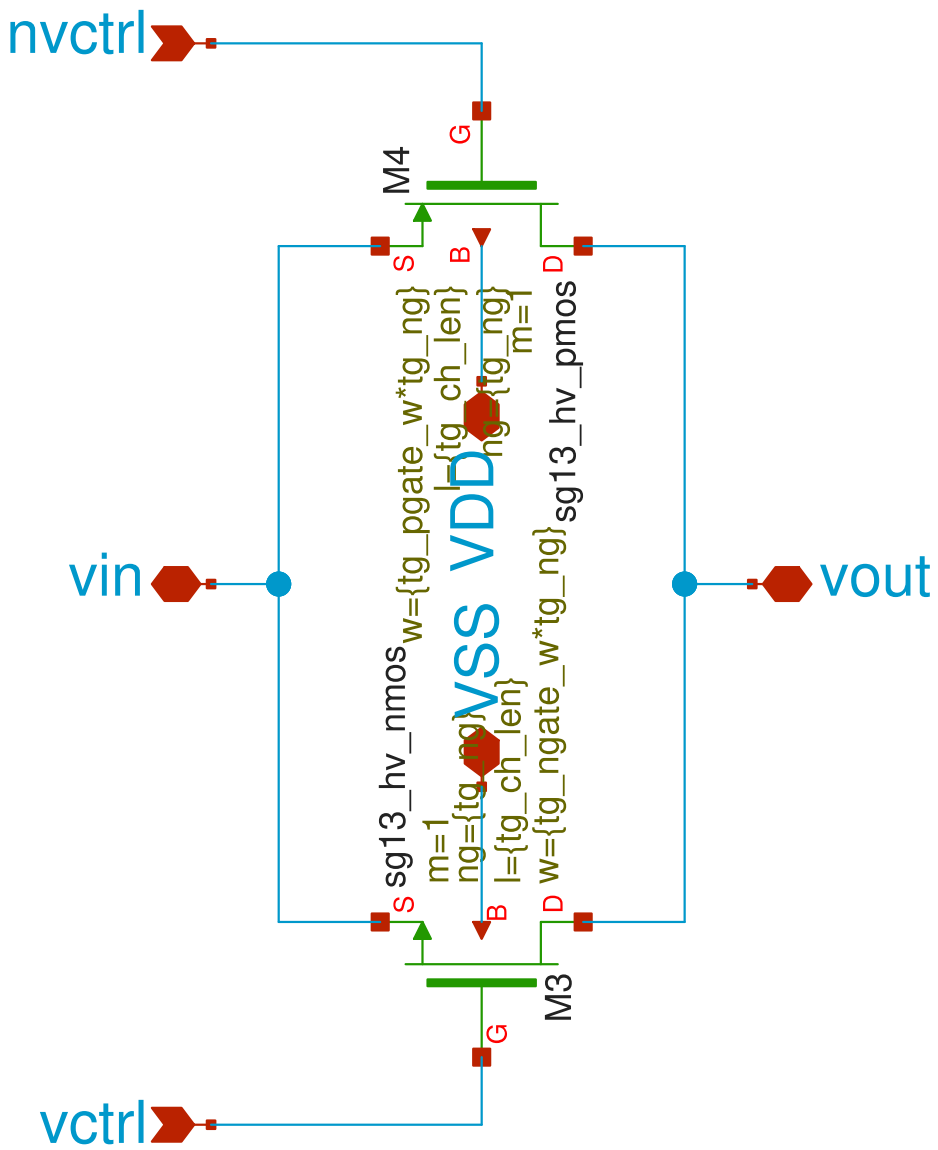
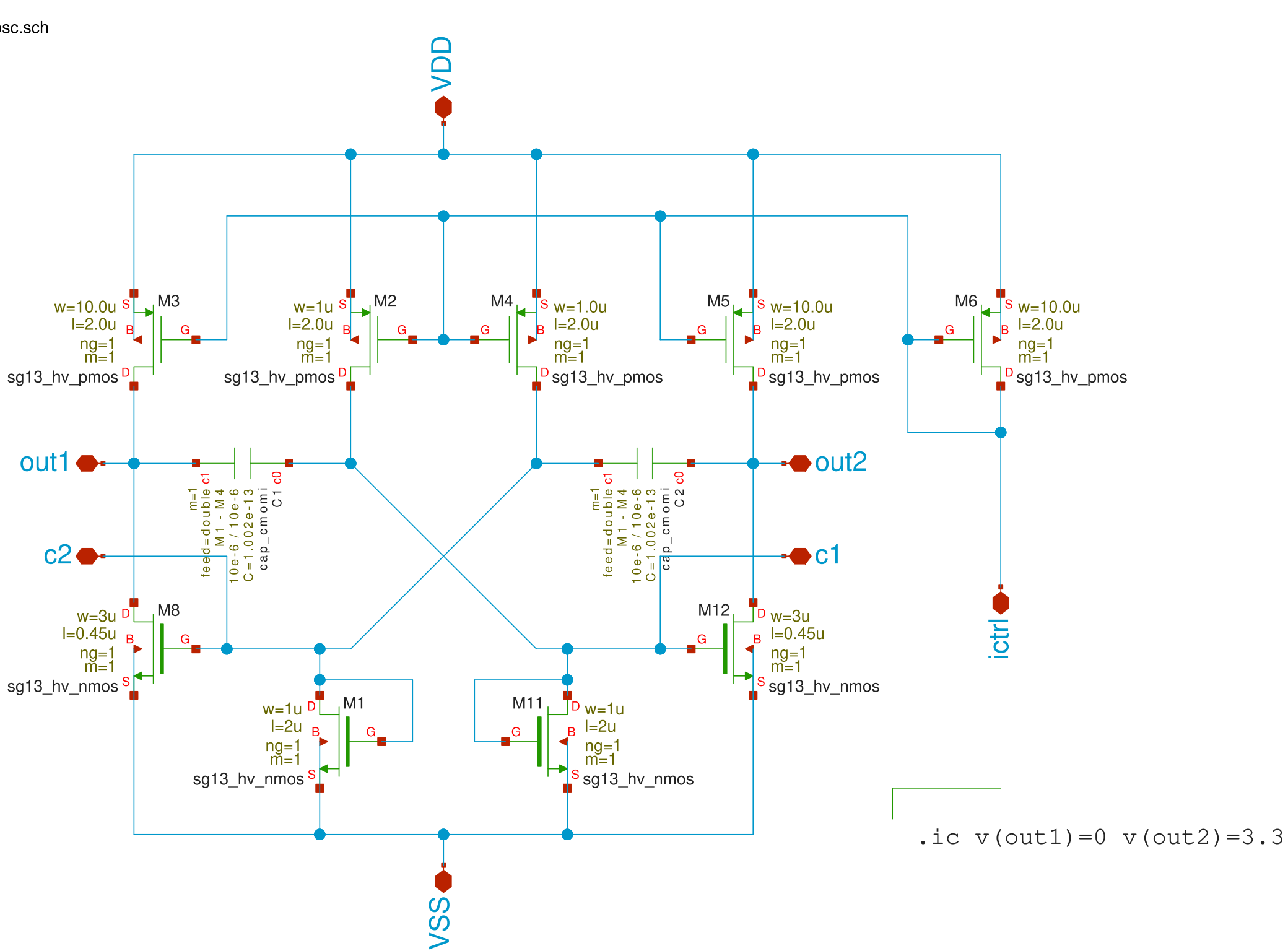
switched_cap_cell.sch



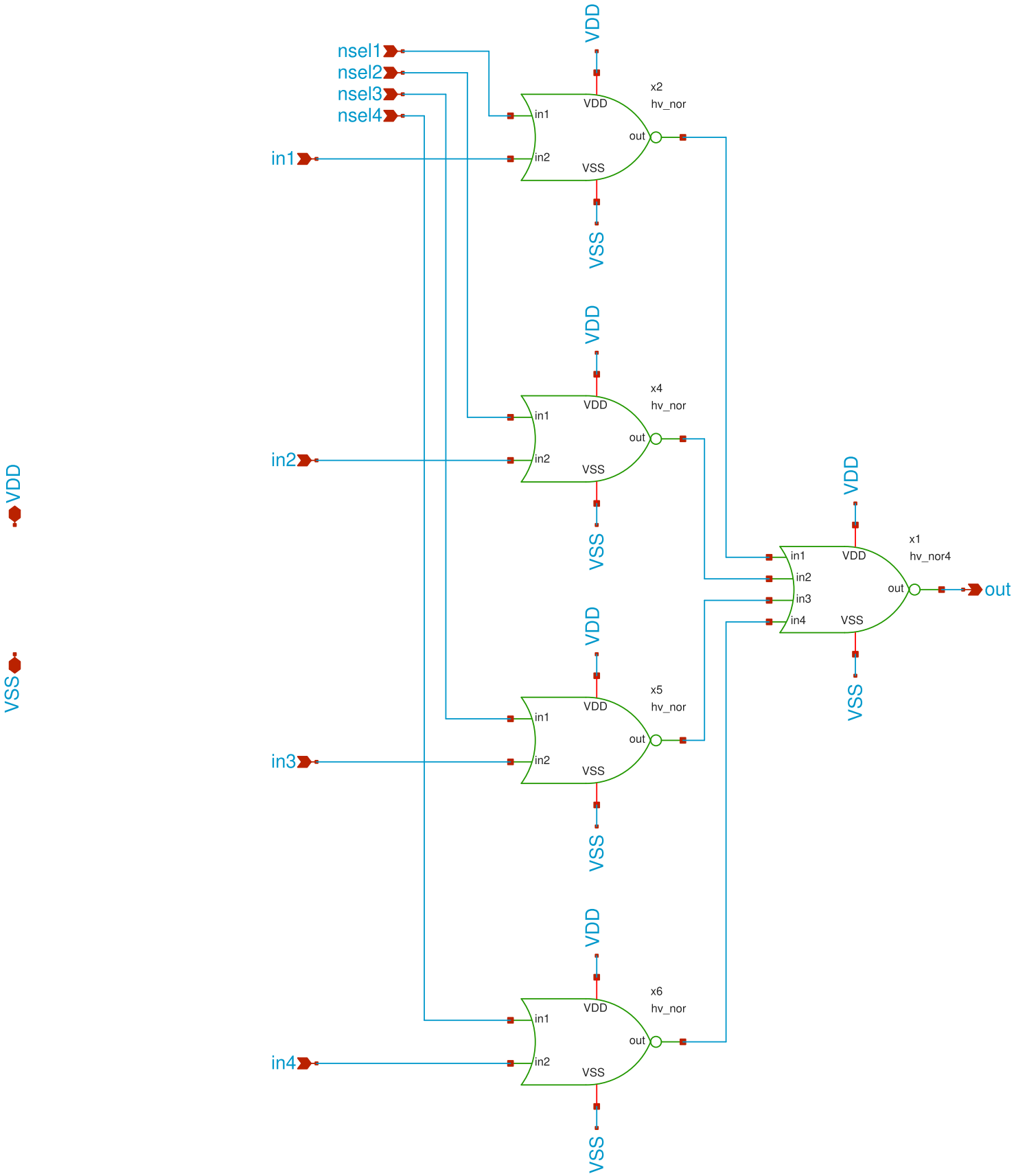


s1

```
.param tg_ch_len=0.45u
.param tg_ngate_w=2u
.param tg_pgate_w=5u
.param tg_ng=20
```


osc.sch
VDD
VSS
out1
out2
c1
c2
ictrl
M1
M2
M3
M4
M5
M6
M8
M11
M12
w=10.0u
l=2.0u
ng=1
m=1
sg13_hv_pmos
w=1u
l=2.0u
w=1.0u
l=2.0u
w=3u
l=0.45u
sg13_hv_nmos
w=1u
l=2u
m=1
feed=double
M1 - M4
10e-6 / 10e-6
C=1.002e-13
cap_cmomi
C1 c0
C2 c0
.ic v(out1)=0 v(out2)=3.3

hv_mux4.sch

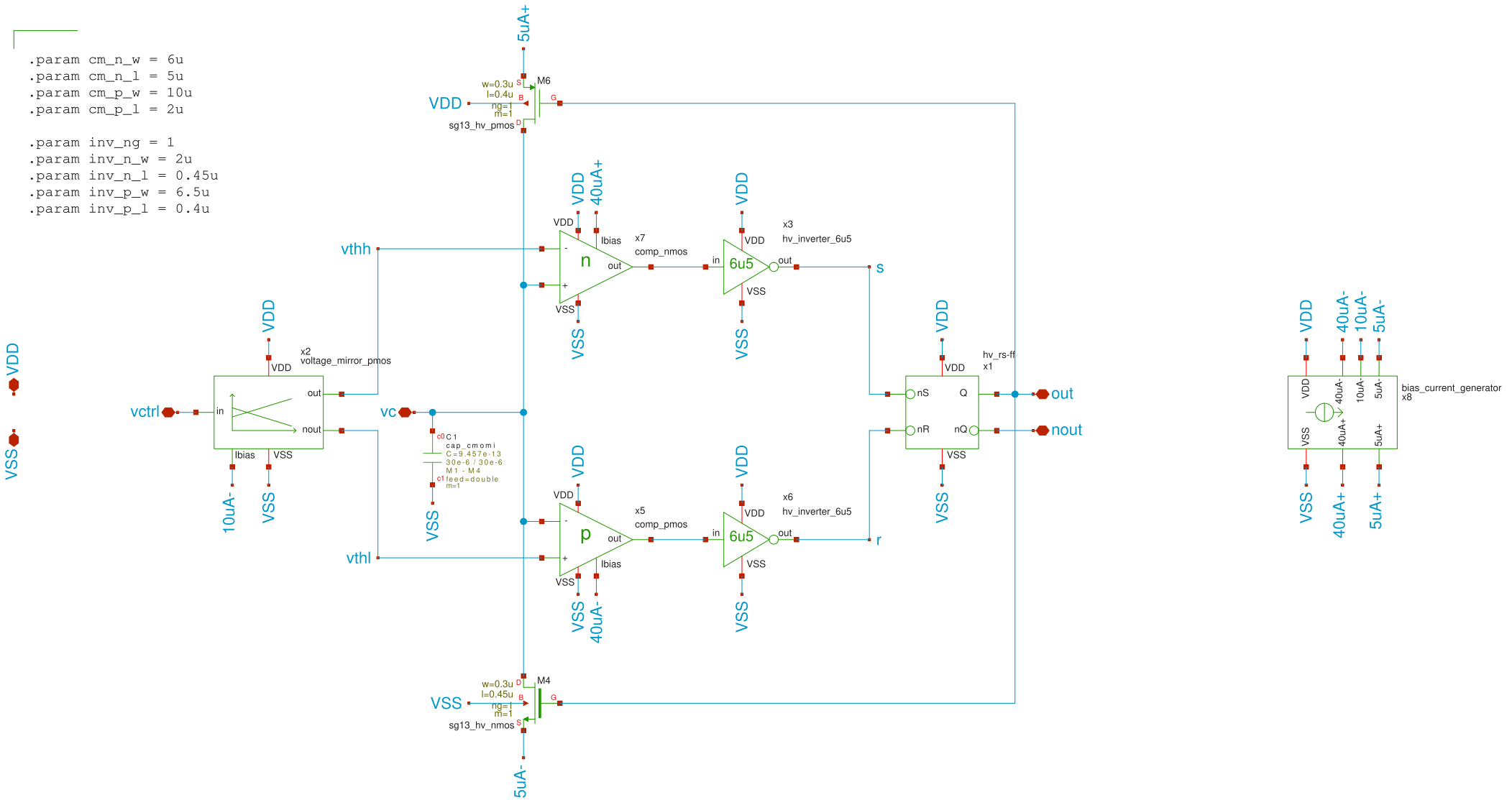

VDD
VSS
nsel1
nsel2
nsel3
nsel4
in1
in2
in3
in4
x2
hv_nor
x4
hv_nor
x5
hv_nor
x6
hv_nor
x1
hv_nor4
in1
in2
in3
in4
out
out

vco.sch

```
.ic v(vc)=1.65

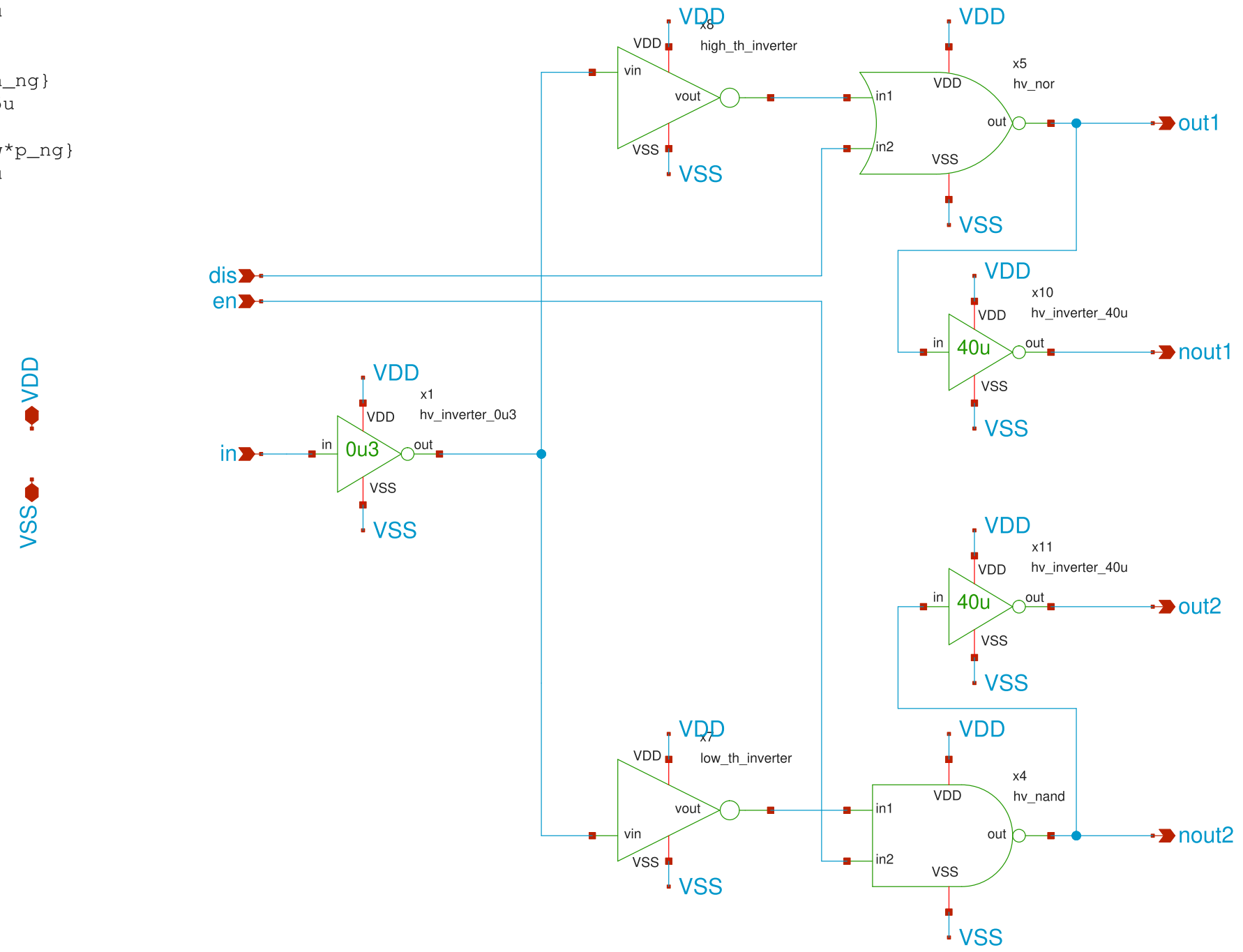
.param cm_n_w = 6u
.param cm_n_l = 5u
.param cm_p_w = 10u
.param cm_p_l = 2u

.param inv_ng = 1
.param inv_n_w = 2u
.param inv_n_l = 0.45u
.param inv_p_w = 6.5u
.param inv_p_l = 0.4u
```

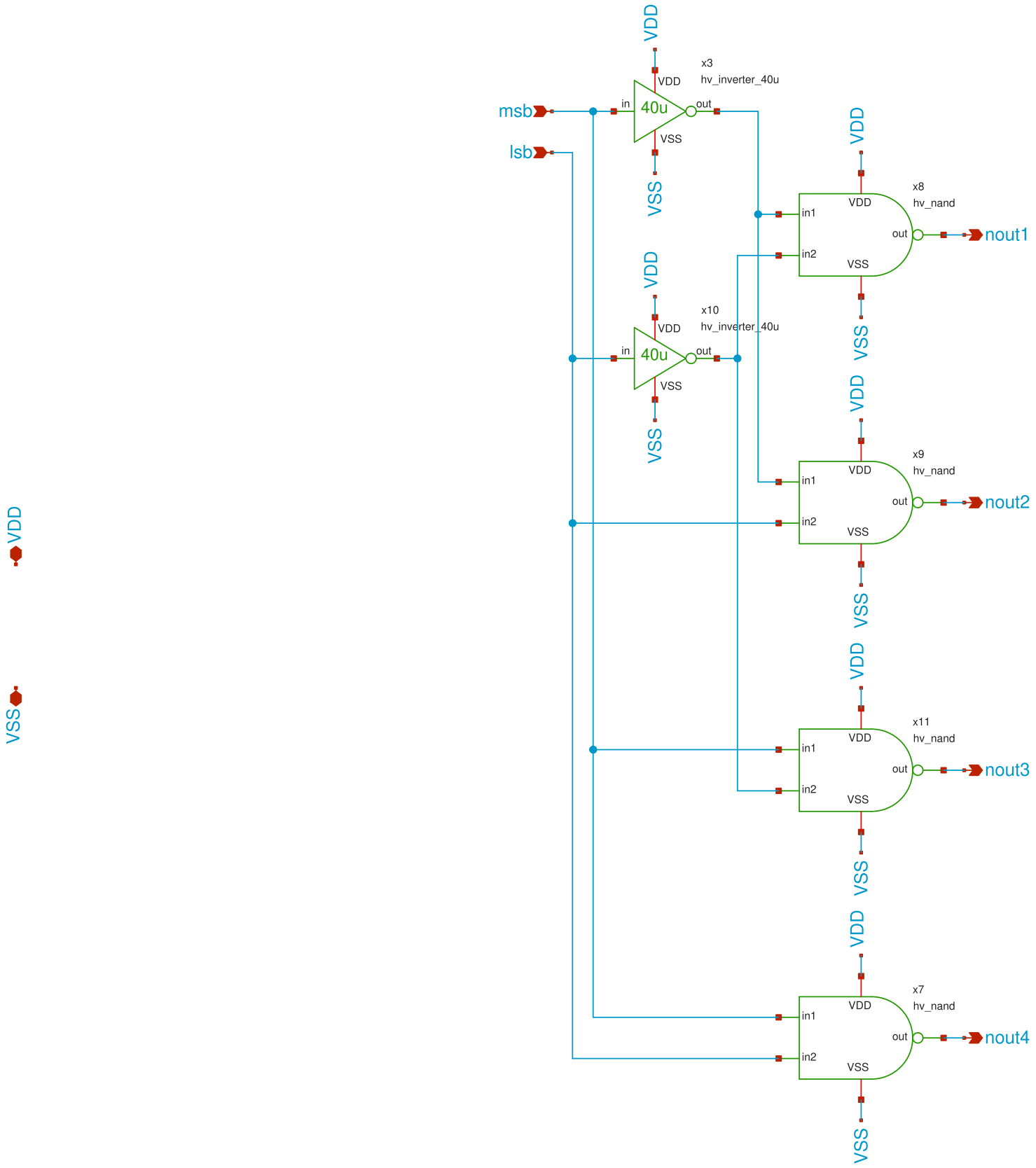


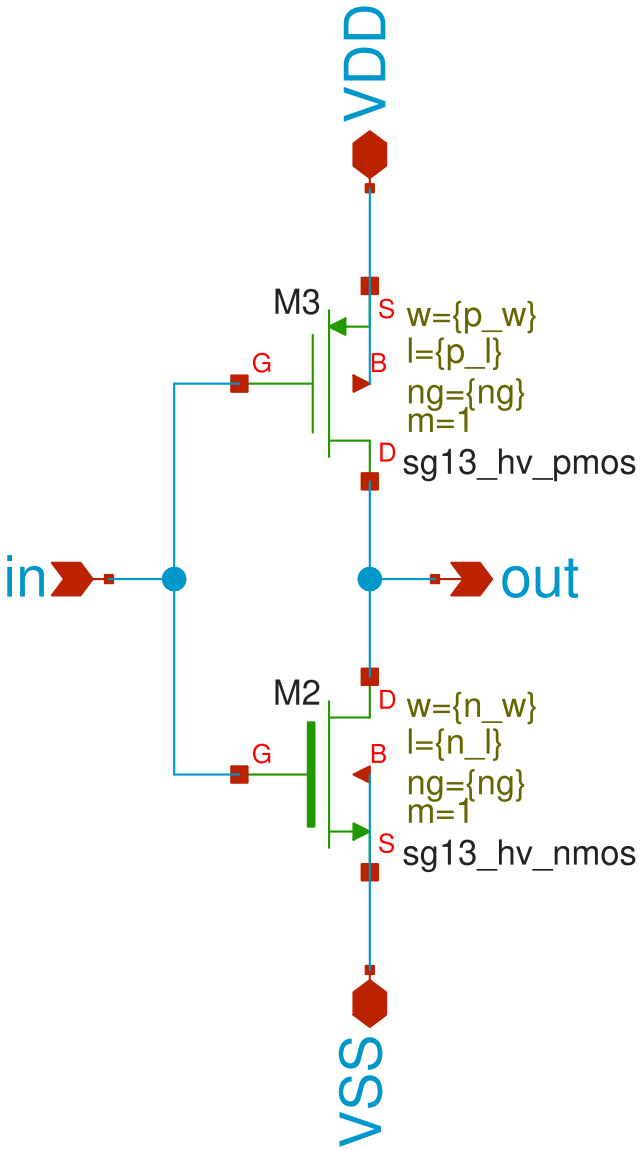
pulse_shaper_en.sch

```
.param w    = 0.3u
.param ng   = 1
.param n_ng = ng
.param n_w  = {w*n_ng}
.param n_l  = 0.45u
.param p_ng = ng
.param p_w  = {1*w*p_ng}
.param p_l  = 0.4u
```



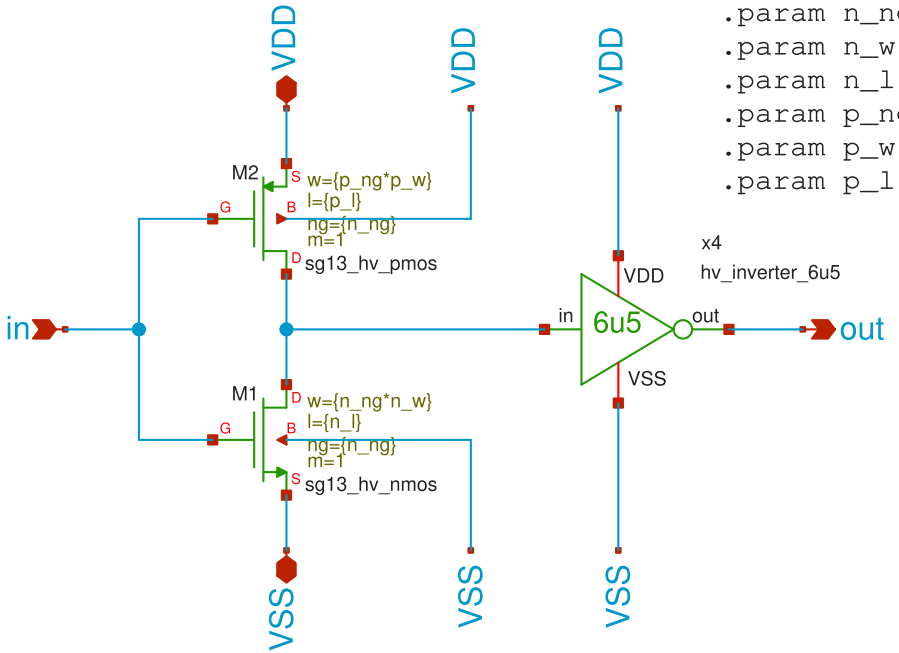
hv_binary_decoder4.sch






```
.param w    = 10u
.param ng   = 4
.param n_ng = ng
.param n_w = {3u*n_ng}
.param n_l = 0.45u
.param p_ng = ng
.param p_w = {10u*p_ng}
.param p_l = 0.4u
```

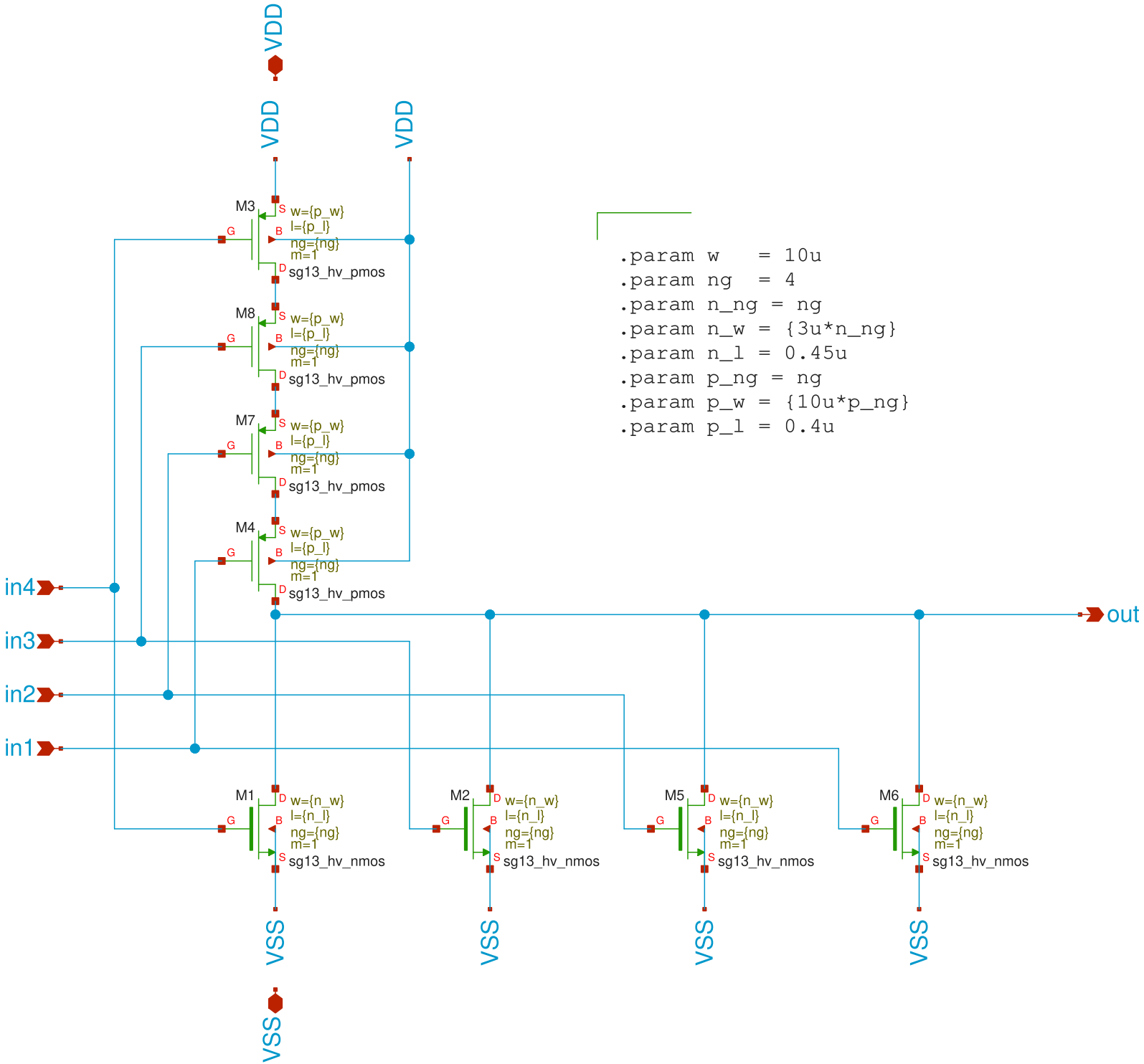
lvl_shift_up.sch



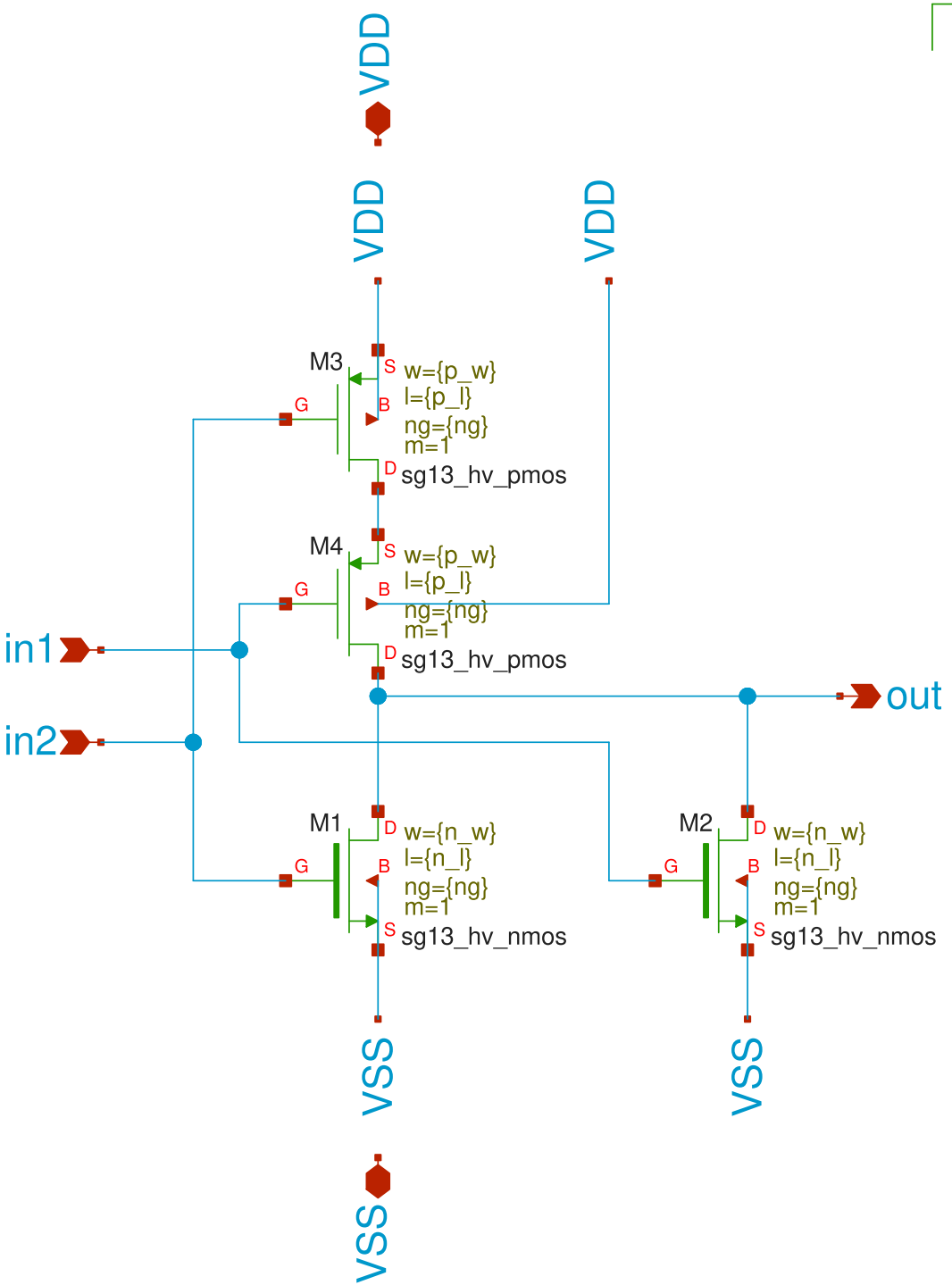

```
.include ../../../macros/switched_capacitor/schematic/xschem/hv_inverter_6u5.sym

.param n_ng = 2
.param n_w = 10u
.param n_l = 0.45u
.param p_ng = 2
.param p_w = 1u
.param p_l = 0.45u
```

hv_nor4.sch

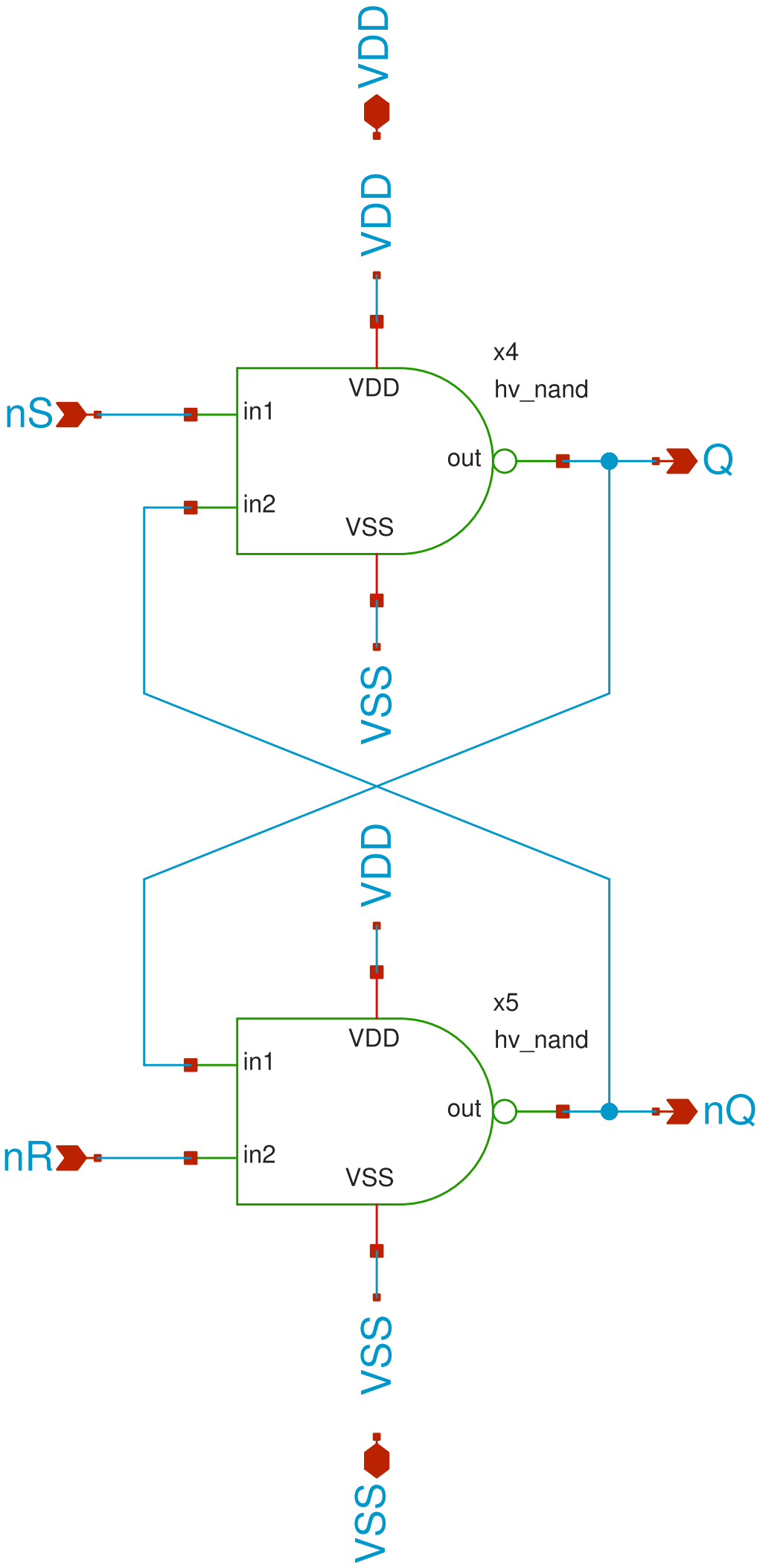

VDD
VDD
VDD
M3
M8
M7
M4
w={p_w}
l={p_l}
ng={ng}
m=1
sg13_hv_pmos
.param w = 10u
.param ng = 4
.param n_ng = ng
.param n_w = {3u*n_ng}
.param n_l = 0.45u
.param p_ng = ng
.param p_w = {10u*p_ng}
.param p_l = 0.4u
in4
in3
in2
in1
out
M1
M2
M5
M6
w={n_w}
l={n_l}
ng={ng}
m=1
sg13_hv_nmos
VSS
VSS
VSS
VSS
VSS

hv_nor.sch



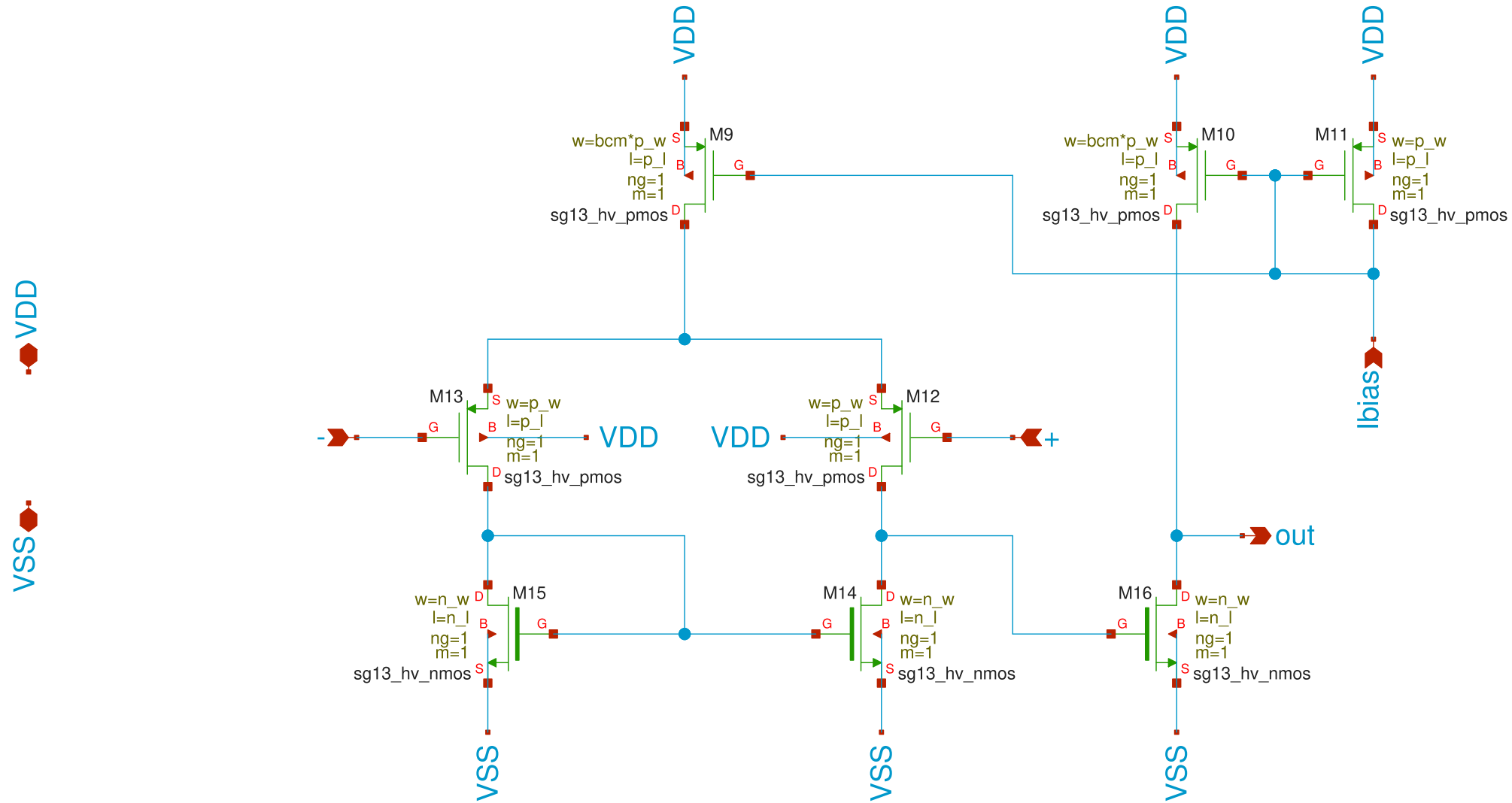

```
.param w    = 10u
.param ng   = 4
.param n_ng = ng
.param n_w = {3u*n_ng}
.param n_l = 0.45u
.param p_ng = ng
.param p_w = {10u*p_ng}
.param p_l = 0.4u
```

hv_rs-ff.sch


VDD
VDD
x4
hv_nand
VDD
nS
in1
out
Q
in2
VSS
VSS
VDD
x5
hv_nand
VDD
in1
out
nQ
nR
in2
VSS
VSS
VSS

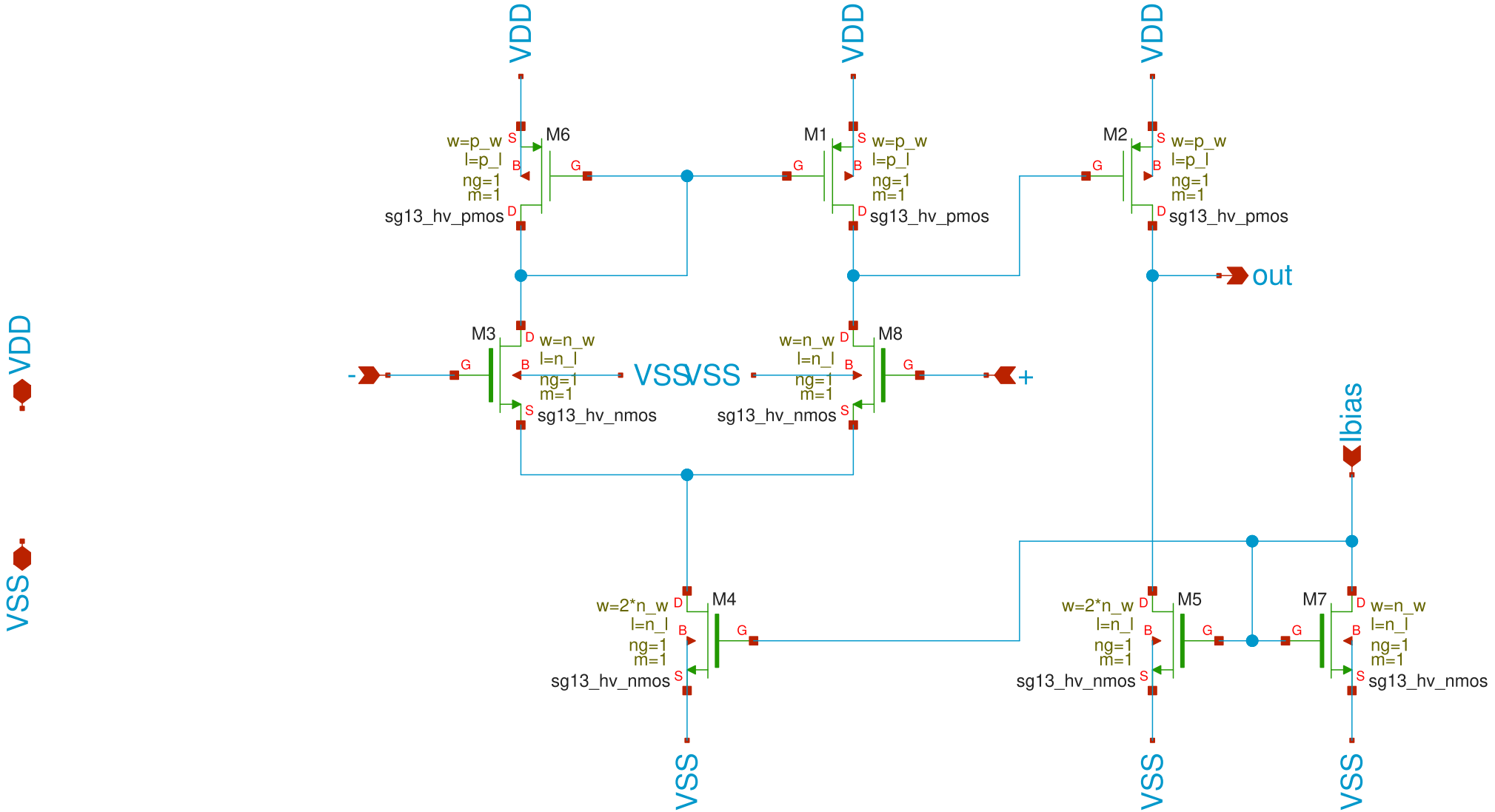
comp_pmos.sch

```
.param n_w = 1.2u
.param n_l = 1.5u
.param p_w = 2u
.param p_l = 0.4u
.param bcm = 1
```

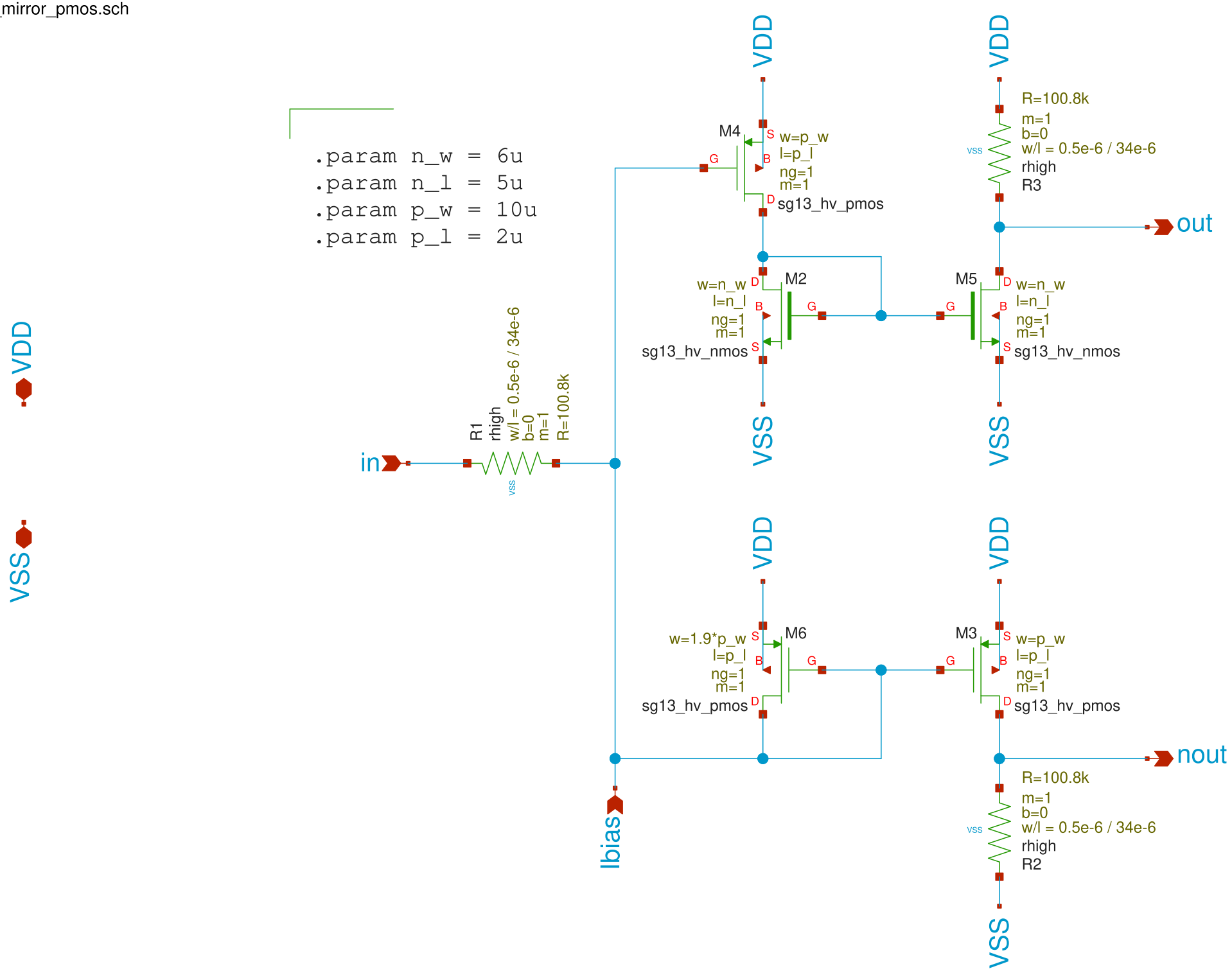


comp_nmos.sch

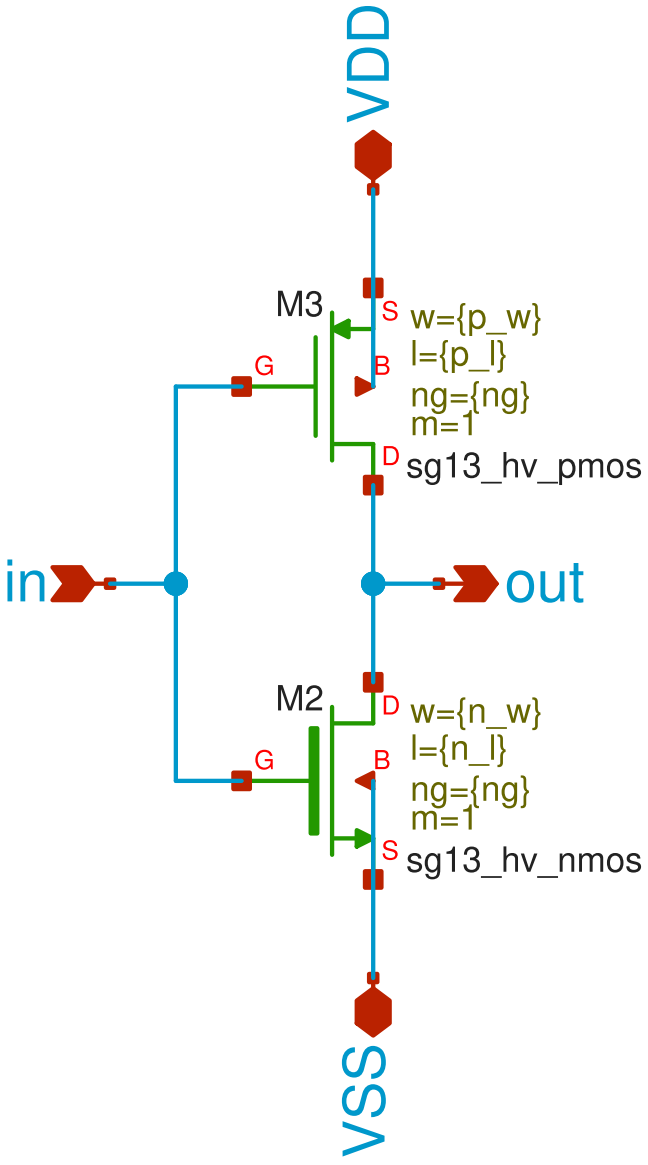
```
.param n_w = 1.2u
.param n_l = 1.5u
.param p_w = 2u
.param p_l = 0.4u
```



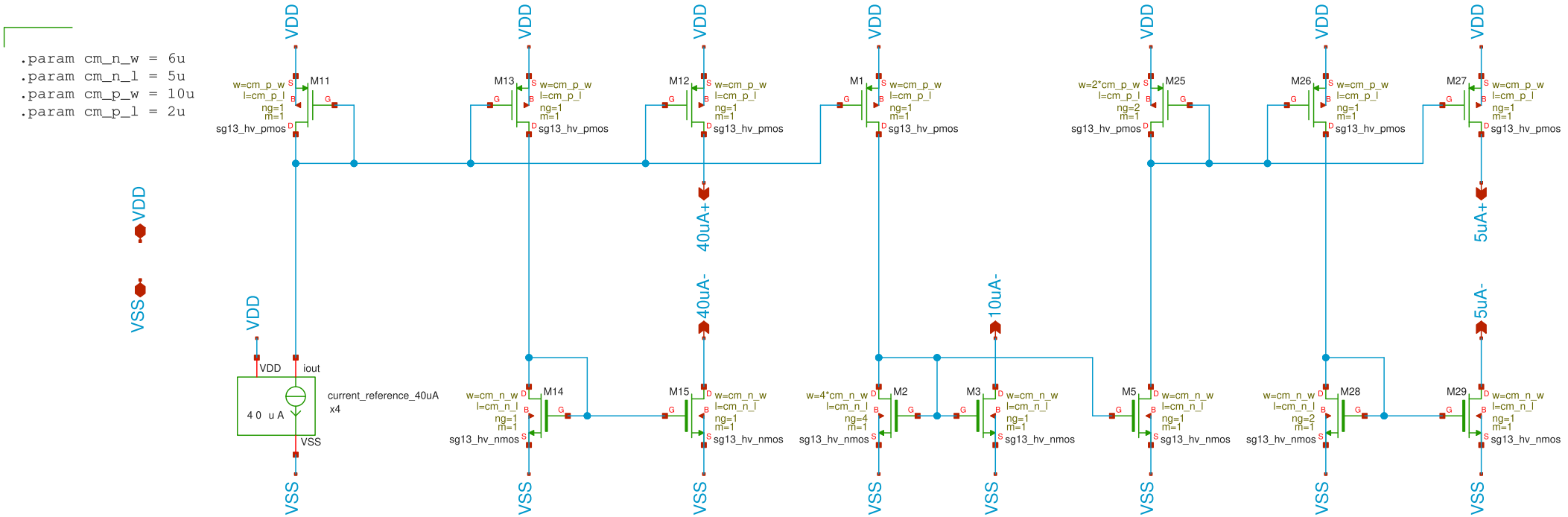
voltage_mirror_pmos.sch



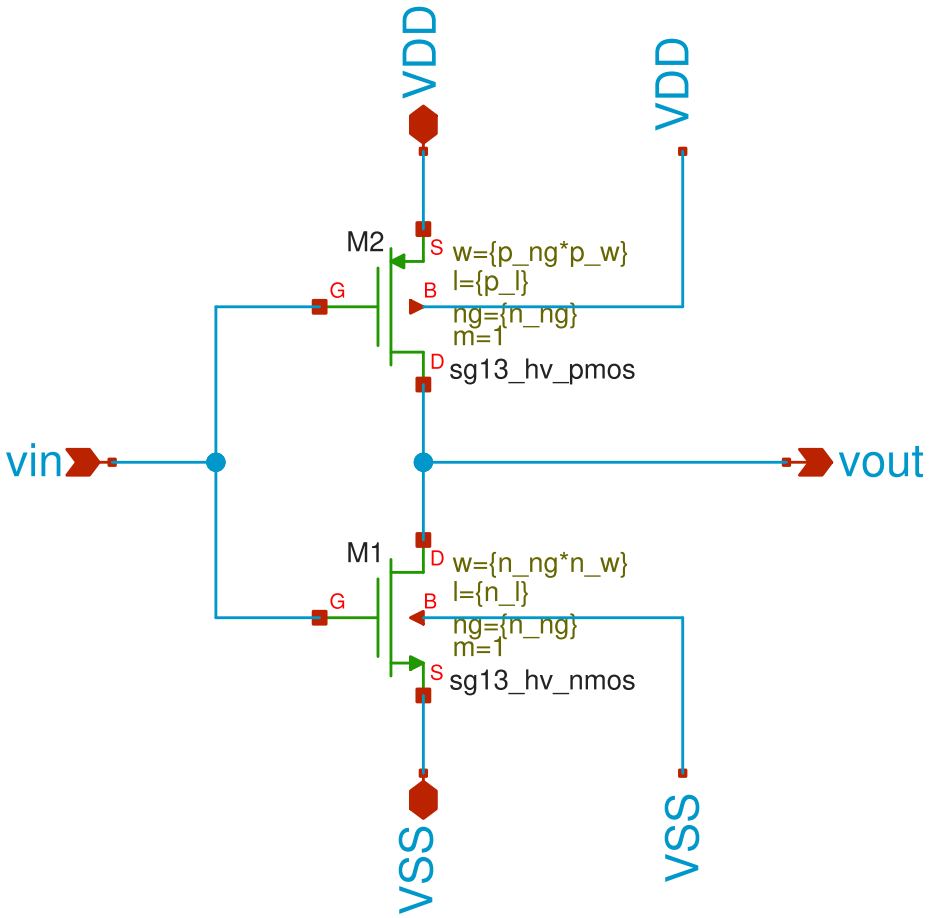
```
.param ng  = 1
.param n_ng = ng
.param n_w = {2u*n_ng}
.param n_l = 0.45u
.param p_ng = ng
.param p_w = {6.5u*p_ng}
.param p_l = 0.4u
```



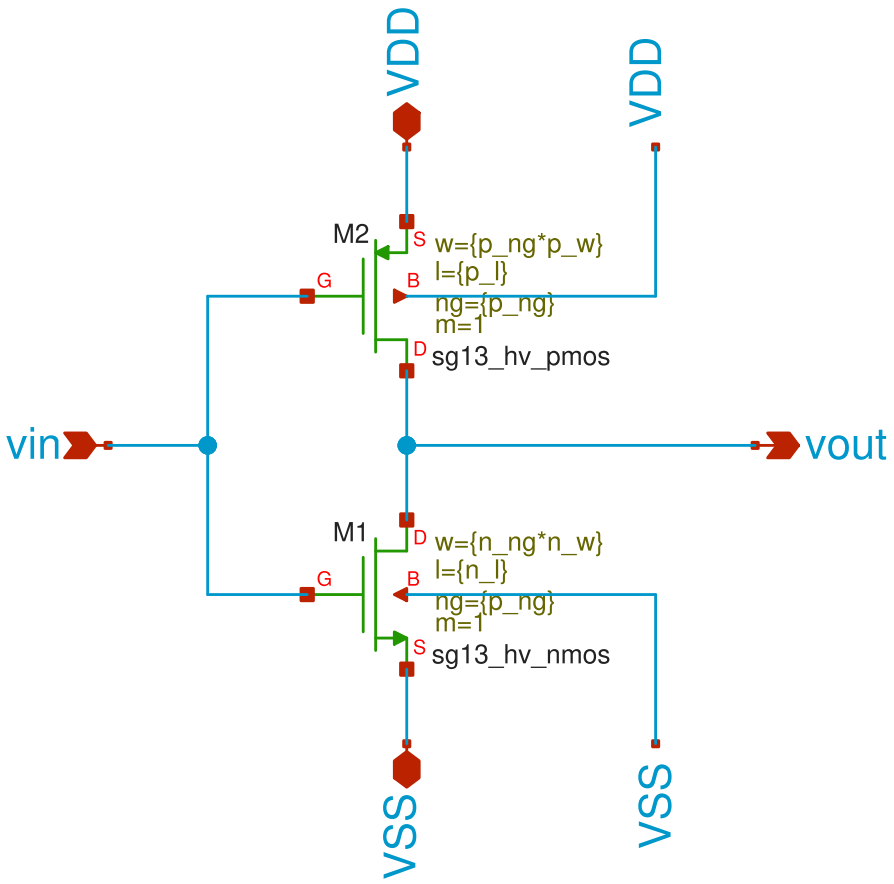
bias_current_generator.sch



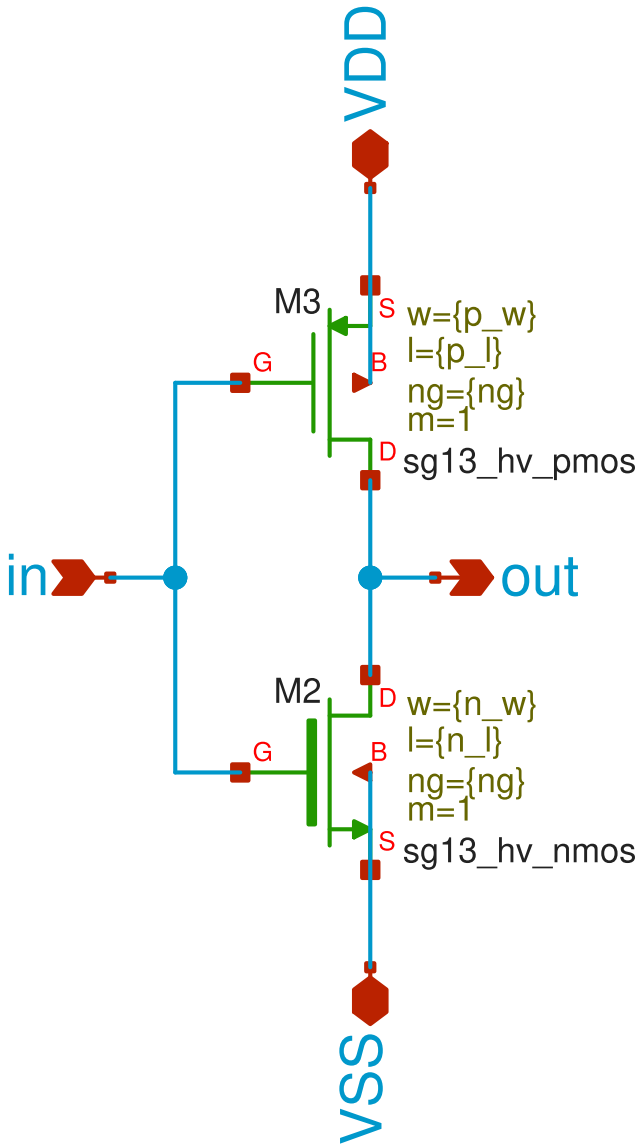
```
.param n_ng = 2
.param n_w = 1u
.param n_l = 0.45u
.param p_ng = 2
.param p_w = 1u
.param p_l = 0.4u
```



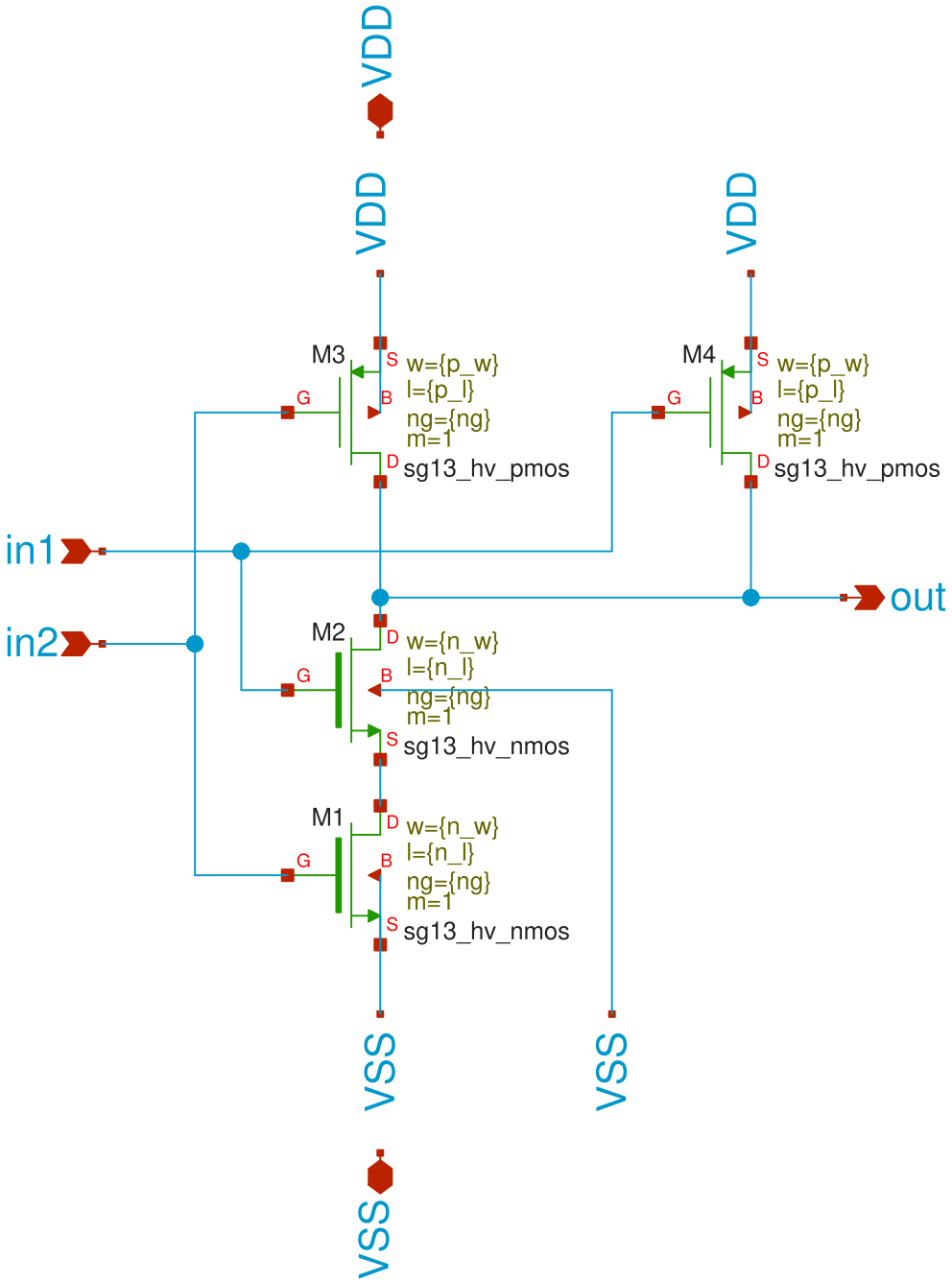
```
.param n_ng = 2
.param n_w = 1u
.param n_l = 0.45u
.param p_ng = 2
.param p_w = 10u
.param p_l = 0.4u
```



```
.param w    = 0.3u
.param ng   = 1
.param n_ng = ng
.param n_w = {w*n_ng}
.param n_l = 0.45u
.param p_ng = ng
.param p_w = {1*w*p_ng}
.param p_l = 0.4u
```



hv_nand.sch




```
.param w    = 10u
.param ng   = 4
.param n_ng = ng
.param n_w = {3u*n_ng}
.param n_l = 0.45u
.param p_ng = ng
.param p_w = {10u*p_ng}
.param p_l = 0.4u
```

current_reference_40uA.sch

